

Profile Summary

Experimental Psychology Ph.D. specializing in Behavioural Neuroscience and advanced neuroimaging. Proven experience with multi-site clinical trial components, developing protocols, and ensuring data quality assurance. Highly skilled in scientific writing and trained in advanced statistical modeling. Accomplished instructor, presenter, speaker adept at disseminating complex findings.

- Skills & Expertise
- Project & Protocol Management

 - Directed the neuroimaging workflow for a Health Canada-regulated, multi-site Randomized Controlled Trial (RCT), including preparing MRI protocol training materials, data acquisition/management, advanced processing, statistics, and manuscript writing.
- Data Analysis

 - Advanced Statistical Modeling (LME Models, Multiple Regression, Permutation Testing), Data Exploitation of Large Scientific Datasets, Statistical Validation of Analytical Results.
- Data Aggregation & Compliance

 - Data Cleaning, Aggregation, Transformation, and Manipulation of Structured/Unstructured Data, Data Integrity & Quality Assurance (QA), Protocol/Legislative Compliance (Health Canada RCT).
- Stakeholder Communication & Advice

 - Communicating Complex Information to Non-Experts (Instructor), Scientific Writing (Peer-Reviewed Publications).

- Proficiencies
- Advanced Analytical Tools

 - R (Statistical Analysis, Data Visualization), Python (Pandas/NumPy for Data Manipulation), SPSS, MATLAB.
- Neuroimaging Tools

 - FSL, SPM, CONN, AFNI.
- Administrative/Research Tools

 - REDCap, SharePoint, OneDrive, Microsoft Office Suite.

- Work Experience
- Research Assistant, *Children's Hospital of Eastern Ontario*

2025 – Present

 - Oversee advanced neuroimaging data collection, regularly reviewing protocols to ensure consistent procedures.
 - Assist technicians with MRI safety procedures and sequence parameters for efficient data acquisition.
 - Perform visual and software-based quality assessments of neuroimaging data, for optimal subsequent analysis.
- Instructor, *University of Ottawa*

2022 – Present

 - Taught the second-year course Biological Foundations of Behaviour to 100+ students in each of four academic terms.
 - Developed, administered, and evaluated exams while providing extensive feedback.
 - Coordinated tasks and develop marking schemes for teaching assistants, while addressing student inquiries.
 - Awarded the William Barry Excellence in Teaching Award in 2024 for my performance as instructor.
- Teaching Assistant, *University of Ottawa*

2020 – 2024

 - Conducted lab tutorials and provided support for 10+ courses in psychology and neuroscience.
 - Collaborated with professors to prepare course materials and grade exams.
- Psychometrist, *Dr. Catherine Gow and Associates*

2019 – 2020

 - Administered a series of tests for cognitive and emotional evaluation of clients seeking neuropsychological assessment.
 - Marked and compiled test results, comparing results to normative data with best practices for confidential data management.
 - Interpreted test results and prepared detailed reports for the on-site neuropsychologist.
- Clinic Support Staff, *Ontario Addiction Treatment Centre*

2018

 - Engaged regularly with patients that sought medical services for addiction treatment.
 - Collected and analyzed patient samples for corresponding physicians.
 - Coordinated with various pharmacies in the city to ensure receipt of written prescriptions and required medications.
- Team Lead, *Escape Manor*

2017 – 2019

 - Supervising staff member of the escape room company *Escape Manor*, overseeing employee duties and responsibilities.
 - Hosted hundreds of guests in a variety of escape rooms, including private workplace events, guaranteeing client satisfaction.

Teaching Experience

Instructor, <i>University of Ottawa</i>	
PSY2301 – Biological Foundations of Behaviour (<i>four</i> academic terms)	2022 – Present
• William Barry Excellence in Teaching Award (2024)	
Teaching Assistant, <i>University of Ottawa</i>	
PSY2116 – Quantitative Methods in Psychology II (130hrs)	2024
PSY2301 – Biological Foundations of Behaviour (65hrs)	2024
PSY3301 – Behavioural Neuroscience (65hrs)	2024
PSY3108 – Perception (65hrs)	2023
PSY3108 – Perception (130hrs)	2023
PSY4391 – Neuroscience of Mindfulness (65hrs)	2022
PSY2301 – Biological Foundations of Behaviour (65hrs)	2022
PSY 3307 – Psychometric Methods (65hrs)	2022
PSY3301 – Behavioural Neuroscience (65hrs)	2021
PSY3108 – Perception (130hrs)	2021
PSY1101 – Introduction to Psychology: Foundations (65hrs)	2020

Research Experience

Research Assistant, <i>Children's Hospital of Eastern Ontario</i>	2025 – Present
Research Assistant (Volunteer), <i>York University</i>	2021
Research Assistant (Volunteer), <i>University of Ottawa</i>	2018-2019

Education

Ph.D. in Experimental Psychology, Specialization in Behavioural Neuroscience	2022 – 2025
<i>University of Ottawa</i>	
• GPA: 4.0/4.0	
• My doctoral thesis addressed a critical gap in the scientific literature by identifying mechanisms to which resilience affects neural outcomes following acute pediatric concussion, and the moderating effects of a mindfulness-based intervention on such mechanisms. My work included three peer-reviewed articles from pediatric concussion trials with advanced neuroimaging.	
Honours Bachelor of Arts in Psychology and Minor in Communication	2015 – 2019
<i>University of Ottawa</i>	
• GPA: 3.7/4.0, <i>Magna Cum Laude</i>	

Theses

• Ph.D.	<i>Neural Correlates of Resilience in Pediatric Concussion and the Influence of a Mindfulness-Based Intervention.</i> Supervised by Drs. Andra Smith & Andrée-Anne Ledoux
• B.A.	<i>Emotion Enhancement of Memory in Face Recognition: Encoding, Retrieval, and Gaze Behaviours.</i> Supervised by Drs. Charles Collin & Isabelle Boutet

Scholarships and Funding

• Murray Brown Award: University of Ottawa Pause and Reflect: A Mindfulness Journey, \$500	2025
• Murray Brown Award: University of Ottawa Society for Neuropsychology, \$2,120	2025
• William Barry Excellence in Teaching Award	2024
• Canada Graduate Scholarships (CIHR) – Doctoral competition, \$40,000 per annum	2024 – 2025
• Ontario Graduate Scholarship (OGS) Program, \$15,000 (offer rejected)	2024
• Admission Scholarship – Doctorate, \$9,000 per annum	2021 – 2025
• Ontario Graduate Scholarship (OGS) Program, \$15,000	2022
• Canada Graduate Scholarships (CIHR) – Master's competition, \$17,500	2021
• Admission Scholarship – Master's, \$7,500	2020
• Merit Scholarships of the Faculty of Social Sciences, \$2,000	2017 – 2019

Invited Presentations

• Speaker , <i>University of Ottawa</i> Mindfulness, Stress, and Perception. University of Ottawa Pause and Reflect: A Mindfulness Journey for Psych Students.	2025
• Speaker , <i>University of Ottawa</i> Neuro-Health Awareness Conference.	2025
• Speaker , <i>University of Ottawa Society of Neuropsychology</i> A Day in the Life: Clinical Neuropsychologist & Psychometrist.	2023
• Discussant , <i>Society of Behavioural Medicine</i> Making an impact after the impact”: Optimizing recovery after Traumatic Brain Injury.	2023
• Speaker , <i>Carleton Neuroscience Society</i> My Experience in Academia & Neurophysiology of Pediatric Concussion.	2022
• Guest Lecturer , <i>University of Ottawa</i> Pediatric Concussion and Mindfulness, as part of PSY4391.	2022
• Guest Lecturer , <i>University of Ottawa</i> War and Humanistic Thought, as part of PSY4130.	2022
• Speaker , <i>Canadian Positive Psychology Association</i> Psychological Resilience and the Recovery from Pediatric Concussion.	2021

Publications

Peer-Reviewed Journals (Original Studies; $n = 10$)

- **Brown** et al. (submitted). Investigating a Mindfulness-Based Intervention [...] Pediatric Concussion. *NeuroImage: Clinical*.
- **Brown** et al. (submitted). The Impact of an App-Based [...] Functional Connectivity. *Mindfulness*.
- Gazel et al. (submitted). Pain Mechanisms in Psoriatic Arthritis [...] Functional MRI. *Rheumatology*.
- Simas et al. (submitted). Exploring Acute-to-Subacute Neuroanatomical [...] Pediatric Concussion. *Journal of Neurotrauma*.
- **Brown** et al. (2024). Associations Between Changes in Psychological [...] Concussion Recovery. *Brain Connectivity*. (IF: 3.40)
- Sicard et al. (2024). Acceptability, Usability, and Credibility [...] Mixed-Method Study. *Digital Health*. (IF: 3.90)
- O’Kane et al. (2024). Development of Therapeutic Alliance [...] Pediatric Concussion. *JMIR Formative Research*. (IF: 2.20)
- **Brown** et al. (2023). Associations Between Psychological [...] Pediatric Concussion. *Human Brain Mapping*. (IF: 5.40)
- Hatchard et al. (2022). Increased gray matter following mindfulness [...] voxel-based morphometry. *Acta Neurologica Belgica*. (IF: 2.47)
- Boutet et al. (2019). Perceptual Factors Can Drive Age-Related Elevation Of False Recognition. *Frontiers in Psychology*. (IF: 2.22)

Peer-Reviewed Journal (Letter to the Editor)

- **Brown** (2023). Letter to the editor [...] therapy for depression.” *Journal of Affective Disorders*.

Published Abstracts (Refereed Journals; $n = 10$)

- Simas et al. (2025). Exploring Acute-to-Subacute Neurostructural Outcomes in Pediatric Concussion. *Canadian Concussion Network 5th Annual Meeting, Victoria, Canada*.
- **[Poster Presenter] Brown** et al. (2025). Mindfulness-Based Intervention [...] Pediatric Concussion Recovery. *Canadian Concussion Network 5th Annual Meeting, Victoria, Canada*.
- Simas et al. (2024). Exploring Longitudinal Trajectories in Cortical Thickness [...] Pediatric Concussion. *Canadian Concussion Network 4th Annual Meeting, Ottawa, Canada*.
- **[Poster Presenter] Brown** et al. (2024). Pediatric Concussion Alters Associations [...] Default Mode Network. *Canadian Concussion Network 4th Annual Meeting, Ottawa, Canada*.
- **[Poster Presenter] Brown** et al. (2023). Dynamic Associations Between Psychological [...] Injury Recovery. *Society of Behavioural Medicine, Phoenix, USA*.
- **[Poster Presenter] Sicard** et al. (2023). Targeted Mindfulness-Based Intervention [...] Open-Label Pilot Study. *Society of Behavioural Medicine Conference, Phoenix, USA*.
- **[Oral Presenter] Brown** et al. (2023). Associations of psychological resilience [...] pediatric concussion. *International Brain Injury Association, Dublin, Ireland*.
- **[Poster Presenter] Brown** et al. (2022). Psychological Resilience [...] Sub-Acute Phase of Pediatric Concussion. *Canadian Concussion Network 2nd Annual Meeting, Montréal, Canada*.
- Sicard et al. (2022). Acceptability, Usability, and Credibility [...] Mobile App for Pediatric Concussion. *Canadian Concussion Network 2nd Annual Meeting, Montréal, Canada*.
- **[Poster Presenter] Smith** et al. (2022). Can mindfulness help [...] the brain says YES! *Organization for Human Brain Mapping 2022, Virtual*.

Scientific Abstracts (Published in Conference Proceedings; $n = 12$)

- **[Oral Presenter] Brown et al. (2025).** Mindfulness-Based Intervention [...] Adolescents Following Concussion. *Brain Health Research Day, Ottawa, Canada.*
- Simas et al. (2024) Is there a link between Emotional Dysregulation [...] Neural Structural Outcomes? *Neuroscience and Mental Health Trainee Network (NeuMe-TN), Ottawa, Canada.*
- Simas et al. (2024). Is there a link between Emotional Dysregulation [...] Neural Structural Outcomes? *Precision Child and Youth Mental Health Conference, Ottawa, Canada.*
- **[Poster Presenter] Brown et al. (2024).** An Investigation of Psychological Resilience [...] Pediatric Concussion. *Precision Child and Youth Mental Health Conference, Ottawa, Canada.*
- **[Oral Presenter] Brown et al. (2023).** Associations Between Psychological Resilience [...] Injury Recovery. *Brain Health Research Day, Ottawa, Canada.*
- Gazel et al. (2023) Pain Mechanisms in Psoriatic Arthritis [...] in Comparison to Functional MRI. *ACR Convergence, Ottawa, Canada.*
- O’Kane et al. (2022). Development of Therapeutic Alliance and Social Presence [...] for Adolescents with Concussion. *G. F. Strong Rehabilitation Research Day, Virtual.*
- **[Poster Presenter] Brown et al. (2021).** Brain changes following mindfulness-based [...] breast cancer treatment. *Gary Johnson Research Day, Virtual.*
- **[Poster Presenter] Brown et al. (2021).** Mindfulness based stress reduction [...] breast cancer survivors? The brain says yes! *SfN Global Connectome: A Virtual Event, Virtual.*
- Collin et al. (2019). Interactions Between Effects of Emotional Expressions at Encoding and at Retrieval in Face Identification. *Psychonomic Society 2019, Montréal, Canada.*
- **[Poster Presenter] Brown et al. (2019).** Emotion Enhancement of Memory [...] and Gaze Behaviours. *Canadian Society for Brain, Behaviour, and Cognitive Science, Waterloo, Canada.*
- **[Poster Presenter] Brown et al. (2019).** Emotion Enhancement of Memory [...] and Gaze Behaviours. *Honours Research Conference University of Ottawa, Ottawa, Canada.*

Magazine

- **Brown (2021).** Pediatric Concussion: Neurophysiology, Clinical Symptomatology, and Recovery. *Neuropsyched.*

Community Involvement

• Workshop host on MATLAB for the Interdisciplinary Conference in Psychology at the University of Ottawa.	2023
• Peer-reviewer for four manuscripts submitted for publication at high impact journals.	2023 – Present
• Honour’s thesis evaluation reviewer for University of Ottawa School of Psychology.	2021
• Conference judge for Postdoctoral Research Day, University of Ottawa.	2023
• Conference judge for Brain Health Research Day, University of Ottawa.	2021
• Student ambassador for the Canadian Positive Psychology Association.	2021
• Co-Founder of the reputable graduate organization, the uOttawa Society for Neuropsychology.	2023 – 2025
• Project lead for uOttawa initiative on mindfulness, stress, and perception by directing collaborative seminars.	2025

Award

• William Barry Excellence in Teaching Award	2024
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